SOLUTION DESIGN DOCUMENT (SDD)

Project Title: Machine Learning for Credit Risk Modeling

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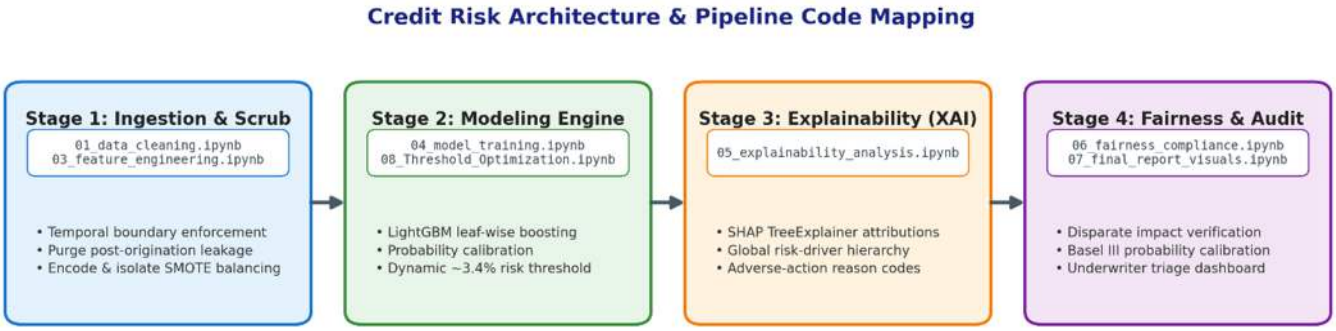
EXECUTIVE BUSINESS CASE & OBJECTIVES

Traditional credit scorecards struggle with complex borrower relationships, while advanced AI algorithms are frequently rejected by financial regulators as opaque "black boxes". This project delivers a machine learning credit underwriting solution that resolves this trade-off. Using loan data from the OpenIntro schema (10,000 borrowers, 1.8% historical default rate), the system accurately predicts loan default while generating transparent explanations for every lending decision.

The primary business objective is to screen incoming loan applicants at origination, tripling default detection efficiency over random selection while strictly adhering to Basel III risk soundness, SR 11-7 model governance, and ECOA fair lending guidelines.

SOLUTION ARCHITECTURE & DATA FLOW

The solution processes incoming loan files through a four-stage pipeline: **Data Cleansing & Leakage Prevention → Balancing & Model Execution → Explainability Layer → Fairness & Operational Triage**.



CODE REPOSITORY & FUNCTIONAL MODULE MAPPING

To ensure auditability, each phase of the workflow corresponds to a distinct module within the codebase:

Code Module / Notebook	Operational & Business Purpose	Key Deliverables / Safeguards
01_data_cleaning.ipynb	**Data Ingestion & Integrity Auditing:** Standardizes raw loan files, maps target outcomes (0 = Paid/Current, 1 = Default), and imputes missing credit bureau fields.	Fills joint metrics, tags clean records, and handles unrecorded delinquency history.

02_exploratory_data_analysis.ipynb	**Portfolio Risk Profiling:** Verifies baseline economic assumptions across credit grades and measures default imbalance.	Confirms monotonic risk graduation across Grades A–F; identifies severe 1.8% default skew.
03_feature_engineering.ipynb	**Target Leakage Prevention & Resampling:** Enforces strict origination boundaries by dropping post-disbursement metrics (paid_principal, paid_total).	Splits data 70/30; isolates SMOTE oversampling to the training set so evaluation remains pristine.
04_model_training.ipynb	**Predictive Engine Benchmarking:** Trains and benchmarks Logistic Regression, Random Forest, XGBoost, and LightGBM.	Identifies LightGBM as the optimal engine, avoiding trivial majority-class collapse.
05_explainability_analysis.ipynb	**Regulatory Explainability (SHAP):** Converts complex tree splits into readable margin contributions for compliance audits.	Produces global risk rankings and borrower-level force plots for Adverse Action disclosure.
06_fairness_compliance.ipynb	**Fair Lending & Calibration Audit:** Evaluates False Positive Rate parity across income brackets to detect disparate impact.	Confirms favorable FPR for low-income applicants (0.07% vs. 0.13%); generates Basel III reliability curves.
07_final_report_visuals.ipynb	**Reporting & Performance Synthesis:** Consolidates model confusion matrices, metric tables, and operational charts.	Provides executive-level reporting artifacts and native split importance summaries.
08_Threshold_Optimization.ipynb	**Operational Decision Calibration:** Optimizes the classification cut-off using Precision-Recall trade-offs.	Establishes the ~3.4% decision boundary, maximizing default capture (32.1% Recall).

OPERATIONAL RISK CONTROLS & GOVERNANCE

- **Target Leakage Safeguard:** Variables tracking post-origination behavior were removed; models trained with leaked artifacts showed an artificially inflated 0.83 AUC, whereas the deployed leakage-free baseline produces a sound 0.66 AUC.
- **Operational Triage Function:** At the 3.4% decision boundary, the model captures ~1 in 3 defaults (32.1% Recall) with a 3.9x concentration over baseline prevalence. Flagged loans are directed to human underwriters alongside SHAP explanation summaries rather than being autonomously denied.
- **Disparate Impact Monitoring:** Evaluated against economic proxies, verifying that low-income borrowers face no algorithmic bias in false rejection rates.